OXFORD AQA
INTERNATIONAL QUALIFICATIONS

INTERNATIONAL GCSE

PHYSICS

9203/2

Paper 2

Mark scheme

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Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from oxfordaqa.com

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 2 with a small amount of level 3 material it would be placed in level 2 but be awarded a mark near the top of the level because of the level 3 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:
- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives, level of demand and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

2.1 In a list of acceptable answers where more than one mark is available 'any **two** from' is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.

2.2 A bold **and** is used to indicate that both parts of the answer are required to award the mark.

2.3 Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.

2.4 Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that 'right + wrong = wrong'.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	**Response**	**Marks awarded**
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	**Response**	**Marks awarded**
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols/formulae

If a student writes a chemical symbol/formula instead of a required chemical name, full credit can be given if the symbol/formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(…..) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, ‘allow’ is used to indicate creditworthy alternative answers.

3.9 Ignore

Ignore is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do <u>not</u> accept

Do **not** accept means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	14 thousand million years		1	AO1 3.8.3d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	(the) Big Bang (theory)		1	AO1 3.8.3d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	red shift		1	AO1 3.8.3b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	the further away a galaxy is from the Earth, the faster the galaxy is moving the space that makes up the universe is expanding		1 1	AO1 3.8.3b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	any **one** from: • measurements were not accurate due to technology (in 1929) • the light from distant galaxies is very faint • the distances are very big • distance is difficult to measure	 allow the universe is very big	1	AO3 3.8.3b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.6	C – cosmic M – microwave B – background		1	AO1 3.8.3c

Total Question 1		**7**

Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	protons **and** neutrons	either order	1	AO1 3.7.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	protons **and** electrons	either order	1	AO1 3.7.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	11 13	must be in this order	1 1	AO1 AO2 3.7.1f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	about one metre		1	AO1 3.7.2g

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	limit exposure time wear a lead-lined apron		1 1	AO3 3.7.2i

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.6	the number of sodium-24 atoms decreases		1	AO3 3.7.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.7	the time it takes for the number of nuclei of the isotope in a sample to halve **or** the time it takes for the count rate from a sample containing the isotope to fall to half its initial level		1	AO1 3.7.2h

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.8	half-life = 15 (hours)	allow 14 to 16	1	AO2 3.7.2h
	time = 15 × 4	allow their half-life × 4	1	
	time = (60 hours)	allow an answer consistent with their half-life	1	
	OR count rate = $\frac{150}{16}$ (1) count rate = 9.375 (1)			
	time = 60 (hours) (1)	allow an answer consistent with their count rate		

Total Question 2		**12**

Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	8760		1	AO2 3.6.5f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	$P = \frac{250 \times 20}{100}$		1	AO2 3.6.5f
	$P = 50$		1	
	$P = 0.050$ (kW)	allow a correct conversion of their incorrectly calculated value of P	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	cost = 8760 × 0.050 × 0.40		1	AO2 3.6.5f
	$175.20		1	
		allow ecf from Question **03.1** and Question **03.2**		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	more efficient devices waste a lower proportion of energy	allow waste less energy / power	1	AO3 3.6.5f
	uses less electricity		1	

Total Question 3		**8**

Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	$640 = \frac{1}{2} \times 20 \times v^2$		1	AO2 3.2.1e
	$v^2 = \frac{2 \times 640}{20}$ **or** $v = \sqrt{\frac{2 \times 640}{20}}$		1	
	v = 8.0 m/s		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	the kinetic energy of the ball becomes zero		1	AO1 AO2 AO3 3.2.2a
	energy is transferred to kinetic energy of the air	allow transferred to elastic potential energy (of the ball)	1	
	causing heating of the air / surroundings / ball	ignore sound	1	

Total Question 4		**6**

Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	mark the position of the ray where it enters block		1	AO4 3.3.5e
	mark the position where the ray of light hits the plane surface of the glass block	allow mark the position where the ray leaves the glass block	1	
	join the points with a pencil and ruler (when the block is removed)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	the ray is incident at 90° (to the surface)	allow the angle of incidence is 0° allow the light is incident along a normal	1	AO3 3.3.5a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	light ray reflected		1	AO3 3.3.5g
	angles equal	judge by eye (should be accurate to within ± 5°)	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.4	$2.42 = \dfrac{\sin 28.0}{\sin r}$		1	AO2 3.3.5e
	$r = \sin^{-1}\left(\dfrac{\sin 28.0}{2.42}\right)$	allow $\sin r = \left(\dfrac{\sin 28.0}{2.42}\right)$	1	
	$r = 11.186108673..$		1	
	$r = 11.2$ (°)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.5	n is greater for diamond than glass		1	AO3 3.3.5f
	(so because) $n = \dfrac{1}{\sin c}$	allow $n \propto \dfrac{1}{\sin c}$	1	
	as n increases critical angle decreases	dependent on MP2	1	

Total Question 5		**13**

Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	used a g-clamp to prevent equipment falling over **or** used small masses / weights (1) to reduce the risk of injury if they fell (1)	allow description of g-clamp dependent on MP1 dependent on MP1	1 1	AO4 3.1.8a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	$0.20 \times 40 = 0.40 \times 20$ (so) clockwise moment is equal to anticlockwise moment	allow use of values in m	1 1	AO2 3.1.8b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	$3.92 = m \times 9.8 \times 0.80$ $m = \dfrac{3.92}{9.8 \times 0.80}$ $m = 0.500$ (kg) so 4 slotted masses on the holder	 allow $m = 500$ g	1 1 1 1	$3 \times$ AO2 $1 \times$ AO3 3.2.1d

Total Question 6		**8**

Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	any pair from: • does not release carbon dioxide • so does not contribute to global warming • less fossil fuel extraction • so less environmental damage • can be used with a renewable resource • not using finite resources • cheaper to recharge than refuel (for a given range) • decreased running costs	allow climate change for global warming	2	AO1 3.2.3c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	t = 10 800 (s) **and** E = 270 000 000 (J)	subsequent marks may be awarded if t and/or E are incorrectly/not converted	1	AO2 3.6.5f 3.2.2f
	$P = \frac{270\ 000\ 000}{10\ 800}$		1	
	P = 25 000 (W)	for subsequent marks to be awarded, the correct power equation must have been used	1	
	25 000 = 50 × V		1	
	$V = \frac{25\ 000}{50}$		1	
	V = 500 (V)		1	
	OR			
	t = 10 800 (s) **and** E = 270 000 000 (J) (1)	subsequent marks may be awarded if t and/or E are incorrectly/not converted		
	$50 = \frac{Q}{10\ 800}$ (1)			
	Q = 50 × 10 800 (1)			
	Q = 540 000 (C) (1)	for subsequent marks to be awarded, the correct charge equation must have been used		
	$V = \frac{270\ 000\ 000}{540\ 000}$ (1)			
	V = 500 (V) (1)			

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	(the bigger) battery would have more mass / weight		1	AO3 3.2.2f
	more energy used (to reach a given speed)	allow would need more power (to accelerate the car)	1	
	more energy would be wasted	allow car would be less efficient	1	
		if no other mark awarded, allow **1** mark for the car / battery would be more expensive		

Total Question 7		**11**

Question 8

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.1	**Level 3:** The design / plan would lead to the production of a valid outcome. All key steps are identified and logically sequenced.		5–6	AO4 3.1.1h
	Level 2: The design / plan would not necessarily lead to a valid outcome. Most steps are identified, but the plan is not fully logically sequenced.		3–4	
	Level 1: The design / plan would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.		1–2	
	No relevant content		0	
	Indicative content • hang the spring from a retort stand • measure the original length of the spring with a metre rule • attach a pointer to the end of the spring • add 100 g mass to the spring • measure the length of the spring • repeat for different masses • calculate the extension of the spring • length of spring – original length • calculate the force applied to the spring • use w = mg • plot a graph of extension against force • check if the graph is a straight line through the origin some of the indicative content may be gained by a correctly labelled diagram for level **2** there must be a description of varying the force and measuring the extension			

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.2	$e = 0.020 \times 1.5$	allow 0.03	1	4 × AO2 1 × AO1 3.2.1c
	$13.5 = 0.5 \times k \times (0.030)^2$	subsequent marks may be awarded if *e* is incorrect	1	
	$k = \dfrac{13.5}{0.5 \times (0.030^2)}$		1	
	$k = 30\ 000$		1	
	N/m		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.3	(if elastic potential energy increases) extension has increased		1	AO3 3.1.1h
	so tension has increased	dependent on MP1	1	
	so frequency (of sound wave) will increase	dependent on MP1	1	

Total Question 8		**14**

Question 9

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.1	W = 2880 (N)	subsequent marks may be awarded if *W* is incorrectly/not converted	1	AO2 3.1.1e 6–7
	$2880 = 320 \times g$		1	
	$g = \frac{2880}{320}$		1	
	g = 9 (N/kg)	allow 9.0	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.2		allow air resistance for drag throughout		AO1 3.1.6a, b
	initially gravitational force greater than drag so the probe accelerates		1	
	drag increases with speed (so acceleration decreases)		1	
	eventually gravitational force equal to drag so no acceleration	allow resultant force is zero	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.3	drag increases because the surface area increases	allow air resistance for drag	1	AO1 3.1.6a, b
	drag greater than gravitational force	allow resultant force is upwards	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.4	the atmosphere is denser **or** the gravitational field strength (of Venus) is less	allow force for field strength	1	AO3 3.1.6a, b
	so the speed at which the drag is equal to the gravitational force (in the atmosphere of Venus) is less		1	

Total Question 9		11